

Probability

OLIVER WEST

Contents

1	Fundamentals	1
1.1	Combinatorial analysis	3
1.2	Stirling's formula	4
2	Basic applications	9
2.1	Counting surjections	9
2.2	Counting derangements	10
3	Independence and conditional probability	11
4	Probability distributions	16
4.1	Discrete probability distributions	16
4.2	Random variables	17
4.3	Expectation	19
4.4	Moments, variance and covariance	22
4.5	Inequalities on random variables	26
4.6	Joint distribution and conditional expectation	30
5	Random walks	34
6	Probability generating functions	35
6.1	Sum of a random number of random variables	39
7	Branching processes	40
7.1	Extinction probability	41
8	Continuous random variables	43

Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ is a collection of subsets of Ω . We call $\mathcal{F}$ a *σ -algebra* if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;

- if $(A_n)_{n \in \mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ -algebra on Ω . A function

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;
- for any countable disjoint collection $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A .

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ -algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω .

Remark. When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω . However, this framework will become useful when considering uncountable Ω .

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;
- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4 • *Rolling a fair die.* Then $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and

$$\mathbb{P}(A) = \frac{|A|}{6}$$

since all outcomes are equally likely.

- $\Omega = \{\omega_1, \dots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.
- *Picking balls from a bag.* Suppose we have n balls labelled $\{1, \dots, n\}$, and we pick $k \leq n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \dots, n\}, |A| = k\}$, so $|\Omega| = \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.

- *A deck of 52 cards.* Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $|\Omega| = 52!$.

What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.

- *Largest digit.* Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \dots, 9\}^n$, $|\Omega| = 10^n$.

Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then

$$\begin{aligned} \mathbb{P}(A_k) &= \frac{(k+1)^n}{10^n} \\ \implies \mathbb{P}(B_k) &= \mathbb{P}(A_k \setminus A_{k-1}) = \frac{(k+1)^n - k^n}{10^n} \end{aligned}$$

- *Birthday problem.* There are n people. What is the probability that at least 2 of them share the same birthday?

Assume each birthday is equally likely to be one of $\{1, \dots, 365\}$. Then $\Omega = \{1, \dots, 365\}^n$, $|\Omega| = 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^c = \{\text{all birthdays different}\}$.

$$\begin{aligned} \mathbb{P}(A^c) &= \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \\ \implies \mathbb{P}(A) &= 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \end{aligned}$$

Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $|\Omega| = n$. We want to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $|\Omega_i| = n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$\begin{aligned}
 M &= \binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdots \binom{n-(n_1+\cdots+n_{k-1})}{n_k} \\
 &= \frac{n!}{n_1!n_2!\cdots n_k!} \\
 &= \binom{n}{n_1, n_2, \dots, n_k}
 \end{aligned}$$

This is the *multinomial coefficient*.

2. Suppose $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

How many strictly increasing $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.

What about increasing functions? Let's define a bijection

$$\{f : \{1, \dots, k\} \rightarrow \{1, \dots, n\} \text{ inc.}\} \rightarrow \{f : \{1, \dots, k\} \rightarrow \{1, n+k-1\} \text{ strictly inc.}\}$$

which takes f to g with $g(i) = f(i) + i - 1$. Hence we have $\binom{n+k-1}{k}$ such functions.

Definition 1.5 (Asymptotic notation). If (a_n) and (b_n) are sequences, we write $a_n \sim b_n$ if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

§1.2 Stirling's formula

Theorem 1.6 (Stirling)

$$n! \sim n^n \sqrt{2\pi n} e^{-n}$$

as $n \rightarrow \infty$.

Weaker statement. We shall first show a weaker statement, that $\log n! \sim n \log n$.

Define $l_n = \log n! = \log 2 + \cdots + \log n$. We have

$$\log \lfloor x \rfloor \leq \log x \leq \log \lceil x \rceil$$

Integrating from 1 to n gives

$$\begin{aligned}
 \sum_{k=1}^{n-1} \log k &\leq \int_1^n \log x \, dx \leq \sum_{k=1}^n \log k \\
 \implies l_{n-1} &\leq n \log n - n + 1 \leq l_n \\
 \implies n \log n - n + 1 &\leq l_n \leq (n+1) \log(n+1) - (n+1) + 1
 \end{aligned}$$

Now the result follows by dividing through by $n \log n$ and taking the limit. ■

Non-examinable proof of Stirling. For every twice-differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$, for all $a < b$, we have

$$\int_a^b f(x) dx = \frac{f(a) + f(b)}{2}(b - a) - \frac{1}{2} \int_a^b (x - a)(b - x) f''(x) dx$$

obtained by integrating twice by parts. Put $f = \log$, $a = k$, $b = k + 1$ for $k \in \mathbb{N}$, to get

$$\int_k^{k+1} \log x dx = \frac{\log k + \log(k + 1)}{2} + \frac{1}{2} \int_k^{k+1} \frac{(x - k)(k + 1 - x)}{x^2} dx$$

Summing from $k = 1$ to $n - 1$, we have

$$\int_1^n \log x dx = \frac{\log(n - 1)! + \log n!}{2} + \sum_{k=1}^{n-1} a_k$$

where $a_k = \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} dx$. Hence

$$n \log n - n + 1 = \log n! - \frac{\log n}{2} + \sum_{k=1}^{n-1} a_k$$

and so

$$\begin{aligned} \log n! &= n \log n - n + 1 + \frac{\log n}{2} - \sum_{k=1}^{n-1} a_k \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot \exp \left(1 - \sum_{k=1}^{n-1} a_k \right) \end{aligned}$$

Let us now consider

$$\begin{aligned} a_k &= \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} dx \\ &\leq \frac{1}{2} \int_0^1 \frac{1}{2k^2} \int_0^1 x(1-x) dx \\ &= \frac{1}{12k^2} \end{aligned}$$

and so $\sum_{k=1}^{\infty} a_k < \infty$. Now put

$$\begin{aligned} A &= \exp \left(1 - \sum_{k=1}^{\infty} a_k \right) \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot A \cdot \underbrace{\exp \left(\sum_{k=n}^{\infty} a_k \right)}_{\rightarrow 1 \text{ as } n \rightarrow \infty} \\ \implies \frac{n!}{n^n e^{-n} \sqrt{n}} &\rightarrow A \text{ as } n \rightarrow \infty \end{aligned}$$

It remains to show that $A = \sqrt{2\pi}$. Consider

$$2^{-2n} \binom{2n}{n} = 2^{-2n} \frac{(2n)!}{(n!)^2}$$

$$\begin{aligned} &\sim \frac{2^{-2n}(2n)^{2n}e^{-2n}\sqrt{2n}A}{n^{2n}e^{-2n}nA^2} \\ &= \frac{\sqrt{2}}{A\sqrt{n}} \end{aligned}$$

Now, we shall use a different method to show

$$2^{-2n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \text{ as } n \rightarrow \infty$$

Consider

$$I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$$

We have $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and, by integrating by parts, we have

$$I_n = \frac{n-1}{n} I_{n-2} \quad (*)$$

so that

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \dots = \frac{(2n-1)(2n-3)\dots 3 \cdot 1}{(2n)(2n-2)\dots 2} \cdot \frac{\pi}{2} \\ &= \frac{2n(2n-1)(2n-2)\dots 3 \cdot 2 \cdot 1}{2n \cdot 2n \cdot (2n-2) \cdot (2n-2) \dots 2 \cdot 2} \cdot \frac{\pi}{2} \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{2} \cdot 2^{-2n} \binom{2n}{n} \end{aligned}$$

and we can similarly obtain

$$\begin{aligned} I_{2n+1} &= \frac{2n \dots 4 \cdot 2}{(2n+1) \dots 3 \cdot 1} \cdot 1 \\ &= \frac{1}{2n+1} \left(2^{-2n} \binom{2n}{n} \right)^{-1} \end{aligned}$$

Now, we want to show that $\frac{I_{2n}}{I_{2n+1}} \rightarrow 1$ as $n \rightarrow \infty$. By (*), we have

$$\frac{I_n}{I_{n-2}} \rightarrow 1 \text{ as } n \rightarrow \infty$$

But $I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$, so I_n is decreasing in n . This gives

$$\frac{I_{2n}}{I_{2n-2}} \leq \frac{I_{2n}}{I_{2n+1}} \leq \frac{I_{2n-1}}{I_{2n+1}}$$

which shows the result by sandwich. Hence, we get, as $n \rightarrow \infty$,

$$\begin{aligned} &\frac{\frac{\pi}{2} \cdot 2^{-2n} \cdot \binom{2n}{n}}{\frac{1}{2n+1} \cdot \left(2^{-2n} \binom{2n}{n} \right)^{-1}} \rightarrow 1 \\ \implies &\left(2^{-2n} \binom{2n}{n} \right)^2 \sim \frac{1}{\pi n} \end{aligned}$$

■

Proposition 1.7 (Countable subadditivity)

Let (A_n) be a collection of events in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n\right) \leq \sum \mathbb{P}(A_n)$$

Proof. Let $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. By definition, it's immediate that (B_n) is a disjoint collection of sets in $\mathcal{F}$, and $\bigcup B_n = \bigcup A_n$, so

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n\right) &= \mathbb{P}\left(\bigcup B_n\right) = \sum \mathbb{P}(B_n) \\ &\leq \sum \mathbb{P}(A_n) \end{aligned}$$

since $B_n \subseteq A_n$ for all n . ■

Proposition 1.8 (Continuity of probability measures)

Take a sequence of events $(A_n)_{n \in \mathbb{N}}$ such that $A_n \subseteq A_{n+1}$ for all n – we say (A_n) is an *increasing* sequence.

Then $\mathbb{P}(A_n)$ is increasing and converges. In particular,

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcup A_n\right)$$

as $n \rightarrow \infty$.

Proof. Define $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. Then

$$\bigcup_{k=1}^n B_k = A_n$$

and the B_n 's are disjoint. Then

$$\begin{aligned} \mathbb{P}(A_n) &= \mathbb{P}\left(\bigcup_{k=1}^n B_k\right) \\ &= \sum_{k=1}^n \mathbb{P}(B_k) \rightarrow \sum_{k=1}^{\infty} \mathbb{P}(B_k) \end{aligned}$$

Also,

$$\begin{aligned} \sum_{k=1}^{\infty} \mathbb{P}(B_k) &= \mathbb{P}\left(\bigcup B_k\right) \\ &= \mathbb{P}(A_k) \end{aligned}$$

■

Corollary 1.9

Analogously, take a sequence of events (A_n) such that $A_{n+1} \subseteq A_n$ for all n – we say (A_n) is a decreasing sequence. Then

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcap A_n\right)$$

Proof. Take complements and use the previous claim. ■

Proposition 1.10 (Inclusion-exclusion formula for probability measures)

Let $A_1, \dots, A_n \in \mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$

Proof. We proceed by induction.

Base case. ADD

Inductive step. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i \cap A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right) \end{aligned}$$

Applying the inductive hypothesis, we get

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \\ \mathbb{P}\left(\bigcup_{i=1}^{n-1} B_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n) \end{aligned}$$

and subbing this in, we will find that everything works out. ■

Proposition 1.11 (Bonferroni inequalities)

Let $A_1, \dots, A_n$ be events. Then

$$\mathbb{P}\left(\bigcup A_i\right) \begin{cases} \leq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ odd} \\ \geq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ even} \end{cases}$$

where those sums are just truncated expansions of inclusion-exclusion.

Proof. We induct again.

Base case. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$

Inductive step. Suppose r is odd. Then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)$$

By the inductive hypotheses,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &\leq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \\ \mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right) &\geq \sum_{k=1}^{r-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n) \end{aligned}$$

where in the former, we choose to apply the identity for odd r , and in the latter, we apply it for even $r - 1$. Substituting much as for the proof of the inclusion-exclusion formula, we obtain the result. The even case is analogous. ■

Let us now consider the special case where Ω is a finite set and $\mathbb{P}(A) = |A| / |\Omega|$. Then we have

Corollary 1.12 (Inclusion-exclusion principle for sets)

Suppose Ω is finite. Then

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

Proof. Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathbb{P}(A) = |A| / |\Omega|$. Then, apply inclusion-exclusion and cancel $1 / |\Omega|$ from both sides. ■

§2 Basic applications

§2.1 Counting surjections

Define a probability space as follows:

$$\Omega = \{f : \{1, \dots, n\} \rightarrow \{1, \dots, m\}\}$$

$$A = \{f \in \Omega : f \text{ is a surjection}\}$$

What is $|A|$? Let us define a set of events

$$A_i = \{f \in \Omega : i \notin \{f(1), \dots, f(n)\}\}, i \in \{1, \dots, m\}$$

so that

$$\begin{aligned} A &= A_1^c \cap \dots \cap A_m^c = (A_1 \cup \dots \cup A_m)^c \\ \Rightarrow |A| &= |\Omega| - |A_1 \cup \dots \cup A_m| \\ &= |\Omega| - \sum_{k=1}^m (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq m} |A_{i_1} \cap \dots \cap A_{i_k}| \\ &= \underbrace{|\Omega|}_{m^n} - \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} (m-k)^n \\ &= \sum_{k=0}^m (-1)^k \binom{m}{k} (m-k)^n \end{aligned}$$

Lecture 4 – 30/01/26

§2.2 Counting derangements

Definition 2.1. A *derangement* is a permutation with no fixed points.

Let

$$\begin{aligned} \Omega &= \{\text{permutations of } \{1, \dots, n\}\} \\ A &= \{\text{derangements}\} \end{aligned}$$

Let

$$A_i = \{f \in \Omega : f(i) = i\}$$

so that

$$\begin{aligned} A &= A_1^c \cap \dots \cap A_n^c = (A_1 \cup \dots \cup A_n)^c \\ \Rightarrow \mathbb{P}(A) &= 1 - \mathbb{P}\left(\bigcup A_i\right) = 1 - \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \underbrace{\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})}_{\frac{(n-k)!}{n!}} \\ &= 1 - \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} \\ &= 1 - \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!} \\ &= \sum_{k=0}^n \frac{(-1)^k}{k!} \rightarrow e^{-1} \end{aligned}$$

§3 Independence and conditional probability

Definition 3.1 (Independence). Let $A, B \in \mathcal{F}$. We say A and B are *independent* and write $A \perp B$ if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Likewise, a countable collection of events $(A_n)_{n \in \mathbb{N}}$ is said to be independent if, for distinct $i_1, \dots, i_k$,

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k \mathbb{P}(A_{i_j})$$

Remark. Pairwise independence of events does not imply independence.

Example

Toss a fair coin twice. We have

$$\begin{aligned} \Omega &= \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ \mathbb{P}(\{\omega\}) &= \frac{1}{4} \end{aligned}$$

Let

$$\begin{aligned} A &= \{(0, 0), (0, 1)\} \\ B &= \{(0, 0), (1, 0)\} \\ C &= \{(1, 0), (0, 1)\} \end{aligned}$$

We have $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C)$, and furthermore we can check that $A \perp B$, $A \perp C$ and $B \perp C$. But $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\emptyset) = 0$, so they are not independent.

Proposition 3.2

If A is independent of B , then A is independent of B^c .

Proof. We have

$$\begin{aligned} \mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A)(1 - \mathbb{P}(B)) \\ &= \mathbb{P}(A)\mathbb{P}(B^c) \end{aligned}$$

■

Definition 3.3 (Conditional probability). Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. For $A \in \mathcal{F}$ we write the *conditional probability of A given B*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Remark. Notably, if $A \perp B$, then $\mathbb{P}(A | B) = \mathbb{P}(A)$, as we would expect.

Proposition 3.4 (Countable additivity for conditional probability)

Suppose $(A_n)_{n \in \mathbb{N}}$ is a disjoint sequence in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n \mid B\right) = \sum \mathbb{P}(A_n \mid B)$$

Proof. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n \mid B\right) &= \frac{\mathbb{P}(\bigcup A_n \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(\bigcup (A_n \cap B))}{\mathbb{P}(B)} \\ &= \sum \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} \\ &= \sum \mathbb{P}(A_n \mid B) \end{aligned}$$

■

Proposition 3.5 (Law of total probability)

Suppose $(B_n)_{n \in \mathbb{N}}$ is a disjoint collection in $\mathcal{F}$, $\bigcup B_n = \Omega$, and $\mathbb{P}(B_n) > 0$ for every n . Let $A \in \mathcal{F}$, then

$$\mathbb{P}(A) = \sum_n \mathbb{P}(A \mid B_n) \mathbb{P}(B_n)$$

Proof. We have

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}\left(A \cap \left(\bigcup B_n\right)\right) \\ &= \mathbb{P}\left(\bigcup (B_n \cap A)\right) \\ &= \sum \mathbb{P}(B_n \cap A) \\ &= \sum \mathbb{P}(A \mid B_n) \mathbb{P}(B_n) \end{aligned}$$

■

Proposition 3.6 (Bayes' theorem)

Suppose (B_n) are disjoint, $\bigcup B_n = \Omega$ and $\mathbb{P}(B_n) > 0$ for every n . Let $A \in \mathcal{F}$, then

$$\mathbb{P}(B_n \mid A) = \frac{\mathbb{P}(A \mid B_n) \cdot \mathbb{P}(B_n)}{\sum_k \mathbb{P}(A \mid B_k) \mathbb{P}(B_k)}$$

Proof sketch. Use the definitions of $\mathbb{P}(B_n | A)$ and $\mathbb{P}(A | B_n)$, and expand the bottom using the law of total probability. ■

This formula is the basis of so-called Bayesian statistics. We know the probability of $\mathbb{P}(B_n)$ and we have a model which gives us $\mathbb{P}(A | B_n)$. With the formula, we find the *posterior* probability of B_n given that A occurred.

Example (False positives for a rare condition)

Suppose that condition A affects 0.1% of the population. We have a medical test which is positive for 98% of the affected population and 1% of the unaffected population. What is the probability that a random individual who tests positive has the disease?

Put $A = \{\text{individual suffers from } A\}$, $P = \{\text{individual tests positive}\}$. We have $\mathbb{P}(A) = 0.001$, $\mathbb{P}(P | A) = 0.98$, $\mathbb{P}(P | A^c) = 0.01$, so

$$\begin{aligned}\mathbb{P}(A | P) &= \frac{\mathbb{P}(P | A)\mathbb{P}(A)}{\mathbb{P}(P | A)\mathbb{P}(A) + \mathbb{P}(P | A^c)\mathbb{P}(A^c)} \\ &= 0.089\end{aligned}$$

In particular, we can see

$$\mathbb{P}(A | P) = \frac{1}{1 + \frac{\mathbb{P}(P|A^c)\mathbb{P}(A^c)}{\mathbb{P}(P|A)\mathbb{P}(A)}}$$

Since $\mathbb{P}(A^c)$ and $\mathbb{P}(P | A)$ are both very close to one, the relevant ratio is

$$\frac{\mathbb{P}(P | A^c)}{\mathbb{P}(A)}$$

But since $\mathbb{P}(P | A^c) \gg \mathbb{P}(A)$, this causes a very small value for $\mathbb{P}(A | P)$.

Example • What is the probability that I have two boys given that I have two children one of whom is a boy?

Since no further information is given, we take all outcomes to be equally likely. Let

$$BG = \{\text{elder child a boy, younger a girl}\}$$

with GB, BB, GG defined analogously. Then we want

$$\begin{aligned}\mathbb{P}(BB \mid BB \cup BG \cup GB) &= \frac{\mathbb{P}(BB)}{\mathbb{P}(BB \cup BG \cup GB)} \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}\end{aligned}$$

- What about $\mathbb{P}(\text{both boys} \mid \text{eldest a boy})$? This is

$$\mathbb{P}(BB \mid BB \cup BG) = \frac{1}{2}$$

- What about $\mathbb{P}(\text{both boys} \mid \text{one a boy born on a Thursday})$? Denote a boy born on Thursday by T, a boy not born on Thursday by N, and a girl by G. Then we want

$$\begin{aligned}&\mathbb{P}(TT \cup TN \cup NT \mid TT \cup TG \cup GT \cup TN \cup NT) \\ &= \frac{\mathbb{P}(TT \cup TN \cup NT)}{\mathbb{P}(TT \cup TG \cup GT \cup TN \cup NT)} \\ &= \frac{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right) \left(\frac{1}{2} \cdot \frac{1}{7}\right)}{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \cdot \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{1}{2} \cdot \frac{1}{7}\right) + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right) \left(\frac{1}{2} \cdot \frac{1}{7}\right)} \\ &=; \frac{13}{27}\end{aligned}$$

Morally speaking, we're providing information that helps specify the boy in question more, which brings the probability closer to a half.

Example (Simpson's paradox)

Consider the following data:

<i>All applicants</i>	Admitted	Rejected	% Admitted
State	25	25	50%
Indep.	28	22	56%

<i>London schools</i>	Admitted	Rejected	% Admitted
State	15	22	41%
Indep.	5	8	38%

<i>Cambridge schools</i>	Admitted	Rejected	% Admitted
State	10	3	77%
Indep.	23	14	62%

– paradoxically, the state admission rate is higher when the school locations are taken individually, but the independent admission right is higher for the combined data. This phenomenon is called *confounding*, and arises when we aggregate data from disparate populations.

Let

$$\begin{aligned}
 A &= \{\text{individual is admitted}\} \\
 B &= \{\text{individual from London}\} \\
 C &= \{\text{individual from state school}\}
 \end{aligned}$$

Then

$$\begin{aligned}
 \mathbb{P}(A \mid B \cap C) &> \mathbb{P}(A \mid B \cap C^c) \\
 \mathbb{P}(A \mid B^c \cap C) &> \mathbb{P}(A \mid B^c \cap C^c) \\
 \text{but } \mathbb{P}(A \mid C) &< \mathbb{P}(A \mid C^c)
 \end{aligned}$$

Let's examine how this can happen. We have

$$\begin{aligned}
 \mathbb{P}(A \mid C) &= \mathbb{P}(A \cap B \mid C) + \mathbb{P}(A \cap B^c \mid C) \\
 &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \cap B^c \cap C)}{\mathbb{P}(C)} \\
 &= \frac{\mathbb{P}(A \mid B \cap C)\mathbb{P}(B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \mid B^c \cap C)\mathbb{P}(B^c \cap C)}{\mathbb{P}(C)} \\
 &= \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^c \cap C)\mathbb{P}(B^c \mid C) \\
 &> \mathbb{P}(A \mid B \cap C^c)\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^c \cap C^c)\mathbb{P}(B^c \mid C)
 \end{aligned}$$

by the first two inequalities from above. Now, if $\mathbb{P}(B \mid C) = \mathbb{P}(B \mid C^c)$, then we would get $\mathbb{P}(A \mid C) > \mathbb{P}(A \mid C^c)$, which is the intuitive expectation – but here this does not hold, and so we can make the ‘paradox’ hold.

Even more simply, this is just the fact that, if $\frac{a_1}{a_2} > \frac{b_1}{b_2}$ and $\frac{c_1}{c_2} > \frac{d_1}{d_2}$, it does not necessarily hold that $\frac{a_1+c_1}{a_2+c_2} > \frac{b_1+d_1}{b_2+d_2}$.

§4 Probability distributions

§4.1 Discrete probability distributions

Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with Ω either finite or countable – that is, $\Omega = \{\omega_1, \omega_2, \dots\}$. As normal, we take $\mathcal{F} = \mathcal{P}(\Omega)$, and so we need only care about $\mathbb{P}(\{\omega_i\})$ for every i .

Definition 4.1 (Discrete probability distributions). For such a setup, we call the sequence of probabilities $(\mathbb{P}(\{\omega_i\}))_i$ a *discrete probability distribution*. We may also write $p_i = \mathbb{P}(\{\omega_i\})$.

Let us consider some examples of discrete probability distributions.

- ① *Bernoulli distribution*. The Bernoulli distribution models the outcome of a toss of a biased coin. We have $\Omega = \{0, 1\}$, $p_0 = \mathbb{P}(\{0\}) = 1 - p$, $p_1 = \mathbb{P}(\{1\}) = p$ for some p with $p \in [0, 1]$.

- ② *Binomial distribution*. We denote this $\text{Bin}(n, p)$, with $n \in \mathbb{N}$, $p \in [0, 1]$. It models the number of heads seen for n independent tosses of a p -biased coin. We have

$$\Omega = \{0, 1, \dots, n\}$$

$$p_k = \binom{n}{k} p^k (1 - p)^{n-k}$$

- ③ *Multinomial distribution*. We denote this $\text{M}(n, p_1, \dots, p_k)$, where $p_i \geq 0$, $\sum p_i = 1$. Given k boxes and n balls placed in box i with p_i , the multinomial distribution gives the probability of n_l balls in box l for every n . We have

$$\Omega = \left\{ (n_1, \dots, n_k) \in \mathbb{N}^k : \sum_{i=1}^k n_i = n \right\}$$

$$\mathbb{P}(n_1, \dots, n_k) = \binom{n}{n_1} p_1^{n_1} \cdot \binom{n - n_1}{n_2} p_2^{n_2} \cdots \binom{n - n_1 - \dots - n_{k-1}}{n_k} p_k^{n_k}$$

$$= \binom{n}{n_1, n_2, \dots, n_k} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

- ④ *Geometric distribution*. The geometric distribution gives the probability that the first heads achieved on a p -coin is after k tosses:

$$\Omega = \{1, 2, \dots\}$$

$$p_k = (1 - p)^{k-1} p$$

Alternatively, we might want to model the probability that we see k tails before the first heads:

$$\Omega = \{0, 1, \dots\}$$

$$p'_k = (1 - p)^k p$$

- ⑤ *Poisson distribution*. Suppose customers arrive into a shop in $[0, 1]$. Discretise $[0, 1]$ into $\left\{ \left[\frac{i-1}{N}, \frac{i}{N} \right] \right\}$, $i = 1, \dots, N$, $N \in \mathbb{N}$. Independently, in each interval, a customer arrives with probability p , and doesn't arrive with probability $1 - p$.

Then

$$\mathbb{P}(k \text{ customers have arrived}) = \binom{N}{k} p^k (1-p)^{N-k}$$

– in other words, this follows a binomial distribution. Now, take $p = \frac{\lambda}{N}$, so

$$\begin{aligned} \mathbb{P}(k \text{ customers have arrived}) &= \frac{N!}{k!(N-k)!} \left(\frac{\lambda}{N}\right)^k \left(1 - \frac{\lambda}{N}\right)^{N-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{N(N-1)\dots(N-k+1)}{N^k}}_{\rightarrow 1} \underbrace{\left(1 - \frac{\lambda}{N}\right)^{N-k}}_{\rightarrow e^{-\lambda}} \\ &\rightarrow e^{-\lambda} \frac{\lambda^k}{k!} \end{aligned}$$

as $N \rightarrow \infty$. Hence we have

$$\begin{aligned} \Omega &= \{0, 1, 2, \dots\} \\ p_k &= e^{-\lambda} \frac{\lambda^k}{k!} \end{aligned}$$

and we denote this $\text{Poi}(\lambda)$.

§4.2 Random variables

Definition 4.2. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \rightarrow \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for any x . We have the shorthand notation $\{X \leq x\}$ to refer to $\{\omega \in \Omega : X(\omega) \leq x\}$, and $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.

Definition 4.3. Let $A \in \mathcal{F}$. Define the *indicator* of A

$$1_A(\omega) = 1(\omega \in A) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

Since $A \in \mathcal{F}$, 1_A is a random variable.

Definition 4.4. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned} F_X(x) &: \mathbb{R} \rightarrow [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(X \leq x) \end{aligned}$$

Definition 4.5. $X = (X_1, \dots, X_n)$ is called a *random variable in $\mathbb{R}^n$* if

$$(X_1, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$$

is a function such that, for any $x_1, \dots, x_n$,

$$\{\omega : X_1(\omega) \leq x_1, \dots, X_n(\omega) \leq x_n\} \in \mathcal{F}$$

which we denote with the shorthand $\{X_1 \leq x_1, \dots, X_n \leq x_n\}$.

Note that $(X_1, \dots, X_n)$ is a random variable if and only if all of $X_1, \dots, X_n$ are.

Definition 4.6. A random variable X is called *discrete* if it takes values in some countable set S .

Definition 4.7. For a discrete random variable, we are able to write

$$p_x = \mathbb{P}(X = x) = \mathbb{P}(\omega : X(\omega) = x)$$

Then, $(p_x)_{x \in S}$ is called the *probability mass function* of X .

Now we are able to say things like

$$\text{for any } A \subseteq S, \mathbb{P}(X \in A) = \sum_{x \in A} p_x$$

If p_x follows a given distribution, then we say X is a random variable with that distribution or a random variable following that distribution.

Definition 4.8. Let $X_1, \dots, X_n$ be discrete random variables with values in $S_1, \dots, S_n$. They are called *independent* if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \dots \mathbb{P}(X_n = x_n)$$

for all $x_1 \in S_1, \dots, x_n \in S_n$.

Example 4.9

Toss a p -coin N times independently. Then

$$\Omega = \{0, 1\}^N$$

and every $\omega \in \Omega$ is of the form $(\omega_1, \dots, \omega_N)$, where ω_i is the i^{th} outcome. Hence

$$p_\omega = \mathbb{P}(\{\omega\}) = \prod_{k=1}^N p^{\omega_k} (1-p)^{1-\omega_k}$$

Let us also define random variables X_k for each k , satisfying $X_k(\omega) = \omega_k$. Clearly X_k is a Bernoulli r.v. of parameter p . We claim $X_1, \dots, X_N$ are independent: we have

$$\begin{aligned} P(X_1 = x_1, \dots, X_N = x_N) &= \prod_{k=1}^N p^{x_k} (1-p)^{1-x_k} \\ &= \prod_{k=1}^N \mathbb{P}(X_k = x_k) \end{aligned}$$

§4.3 Expectation

Definition 4.10. Suppose Ω is finite or countable, and let $X : \Omega \rightarrow \mathbb{R}$ be a discrete random variable. We say X is *non-negative* if $X \geq 0$ – then we define the *expectation* of X by

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) \cdot X(\omega)$$

We can equivalently let

$$\Omega_X = \{X(\omega) : \omega \in \Omega\}$$

so that

$$\begin{aligned} \mathbb{E}[X] &= \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}(\{\omega\}) \\ &= \sum_{x \in \Omega_X} \sum_{\omega \in \{X=x\}} X(\omega) \mathbb{P}(\{\omega\}) \\ &= \sum_{x \in \Omega_X} \sum_{\omega \in \{X=x\}} x \mathbb{P}(\{\omega\}) \\ &= \sum_{x \in \Omega_X} x \sum_{\omega \in \{X=x\}} \mathbb{P}(\{\omega\}) \\ &= \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x) \end{aligned}$$

or in other words the average of the values taken by X , weighted by their probabilities.

Example 4.11

Suppose $X \sim \text{Bin}(n, p)$. Then

$$\begin{aligned} \mathbb{E}[X] &= \sum_{k=0}^N k \mathbb{P}(X = k) \\ &= \sum_{k=0}^N k \binom{N}{k} p^k (1-p)^{N-k} \\ &= \sum_{k=0}^N k \frac{N!}{(N-k)!k!} p^k (1-p)^{N-k} \\ &= \sum_{k=0}^{N-1} N \cdot \binom{N-1}{k} p^{k+1} (1-p)^{N-k-1} \cdot p \\ &= Np \cdot (p + 1 - p)^{N-1} = Np \end{aligned}$$

Example 4.12

Suppose $X \sim \text{Poi}(\lambda)$. Then

$$\begin{aligned}\mathbb{E}[X] &= \sum_{k=0}^{\infty} k \mathbb{P}(X = k) \\ &= \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \cdot \lambda \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda\end{aligned}$$

Let X be a random variable – then we define

$$\begin{aligned}X^+ &= \max(X, 0) \\ X^- &= \max(-X, 0)\end{aligned}$$

so $X = X^+ - X^-$ and $|X| = X^+ + X^-$. Supposing X is discrete, we can define the expectations of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$, since they are both non-negative.

If at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, we define

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$$

If both are infinite, the expectation of X is undefined.

Also, if $\mathbb{E}[|X|]$ (which we define in the obvious way) is $< \infty$, we say X is integrable. When $\mathbb{E}[X]$ is well-defined, we have

$$\mathbb{E}[X] = \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x)$$

We have some properties of expectations as follows:

- If $X \geq 0$, then $\mathbb{E}[X] \geq 0$.
- If $X \geq 0$ and $\mathbb{E}[X] = 0$, then $\mathbb{P}(X = 0) = 1$.
- If $c \in \mathbb{R}$, then $\mathbb{E}[cX] = c\mathbb{E}[X]$ and $\mathbb{E}[c + X] = c + \mathbb{E}[X]$.
- If X and Y are random variables (and X, Y integrable), then $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
- In the countable case, we take $c_1, \dots, c_n \in \mathbb{R}$ and $X_1, \dots, X_n$ integrable, and then we have

$$\mathbb{E}\left[\sum_{i=1}^n c_i X_i\right] = \sum_{i=1}^n c_i \mathbb{E}[X_i]$$

Let us give a proof of the final identity in the case where Ω is countable. We have

$$\begin{aligned}\mathbb{E}\left[\sum_n X_n\right] &= \sum_{\omega \in \Omega} \left(\sum_n X_n(\omega)\right) \mathbb{P}(\{\omega\}) \\ &= \sum_n \sum_{\omega \in \Omega} X_n(\omega) \mathbb{P}(\{\omega\})\end{aligned}$$

$$= \sum_n \mathbb{E}[X_n]$$

where we can interchange sums by the assumption that the random variables are integrable.

We also have the useful result that, if $X = 1(A)$ for some event A ,

$$\mathbb{E}[X] = \mathbb{P}(A)$$

Proposition 4.13

Let $g : \mathbb{R} \rightarrow \mathbb{R}$, and define the random variable $g(X)$ by

$$g(X)(\omega) = g(X(\omega))$$

Then

$$\mathbb{E}[g(X)] = \sum_{x \in \Omega_X} g(x) \mathbb{P}(X = x)$$

Proof. Put $Y = g(X)$, then

$$\begin{aligned} \{Y = y\} &= \{\omega : Y(\omega) = y\} \\ &= \{\omega : g(X(\omega)) = y\} \\ &= \{\omega : X(\omega) \in g^{-1}(\{y\})\} \\ &= \{X \in g^{-1}(\{y\})\} \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}[Y] &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(Y = y) \\ &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(X \in g^{-1}(\{y\})) \\ &= \sum_{y \in \Omega_Y} y \sum_{x \in g^{-1}(\{y\})} \mathbb{P}(X = x) \\ &= \sum_{y \in \Omega_Y} \sum_{x \in g^{-1}(\{y\})} g(x) \mathbb{P}(X = x) \\ &= \sum_{x \in \Omega_X} g(x) \cdot \mathbb{P}(X = x) \end{aligned}$$

■

Proposition 4.14

Suppose that $X \geq 0$ and takes integer values. Then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \sum_{k=0}^{\infty} \mathbb{P}(X > k)$$

Proof. Since $X \geq 0$ and $X \in \mathbb{N}$, we have

$$X = \sum_{k=1}^{\infty} 1(X \geq k) = \sum_{k=0}^{\infty} 1(X > k)$$

Taking expectations, we obtain

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E} \left[\sum_{k=1}^{\infty} 1(X \geq k) \right] &= \mathbb{E} \left[\sum_{k=0}^{\infty} 1(X > k) \right] \\ &= \sum_{k=1}^{\infty} \mathbb{E}[1(X \geq k)] &= \sum_{k=0}^{\infty} \mathbb{E}[1(X > k)] \\ &= \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) &= \sum_{k=0}^{\infty} \mathbb{P}(X > k) \end{aligned}$$

■

Lecture $n + 1$ – 09/02/26

We present another proof of the inclusion-exclusion formula. First, we should establish some properties about indicator functions:

- $1(A^c) = 1 - 1(A)$;
- $1(A \cap B) = 1(A) \cdot 1(B)$;
- $1(A \cup B) = 1 - (1 - 1(A))(1 - 1(B))$.

More generally,

$$\begin{aligned} 1(A_1 \cup \dots \cup A_n) &= 1 - \prod_{i=1}^n (1 - 1(A_i)) \\ &= \sum_{i=1}^n 1(A_i) - \sum_{1 \leq i_1 < i_2 \leq n} 1(A_{i_1} \cap A_{i_2}) + \dots + (-1)^{n+1} 1(A_1 \cap \dots \cap A_n) \end{aligned}$$

Then, we can take expectations of both sides and apply $\mathbb{E}[1(A)] = \mathbb{P}(A)$ to get

$$\mathbb{P}(A_1 \cup \dots \cup A_n) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$

§4.4 Moments, variance and covariance

Definition 4.15. Let X be a random variable and $r \in \mathbb{N}$. We call $\mathbb{E}[X^r]$, as long as it is well-defined, the r^{th} *moment* of X .

Definition 4.16. The *variance* of X is given by

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Furthermore, the *standard deviation* of X is given by $\sqrt{\text{Var}(X)}$.

We have some properties of variance as follows:

- $\text{Var}(X) \geq 0$, since it is the expectation of a squared quantity;
- If $\text{Var}(X) = 0$, then $\mathbb{P}(X = \mathbb{E}[X]) = 1$;
- If $c \in \mathbb{R}$, then $\text{Var}(cX) = c^2 \text{Var}(X)$ and $\text{Var}(X + c) = \text{Var}(X)$.

Lemma 4.17

The variance of X is also given by

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Proof. We have

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

■

Lemma 4.18

The function $\mathbb{E}[(X - c)^2]$ is minimised when $c = \mathbb{E}[X]$, and at that point is equal to $\text{Var}(X)$.

Proof. We have

$$\begin{aligned} f(c) &= \mathbb{E}[(X - c)^2] \\ &= \mathbb{E}[X^2] - 2c\mathbb{E}[X] + c^2 \\ \implies f'(c) &= -2\mathbb{E}[X] + 2c = 0 \\ \implies c &= \mathbb{E}[X] \end{aligned}$$

■

Example 4.19 ① Suppose $X \sim \text{Bin}(n, p)$. Recall $\mathbb{E}[X] = np$. We have

$$\text{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$$

Then

$$\begin{aligned} \mathbb{E}[X(X-1)] &= \sum_{k=2}^n k(k-1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\ &= (n-1)np^2 \sum_{k=0}^{n-2} \binom{n-2}{k} p^k (1-p)^{n-2-k} \\ &= (n-1)np^2 \end{aligned}$$

so that $\text{Var}(X) = (n-1)np^2 + np - n^2p^2 = np(1-p)$.

② Suppose $X \sim \text{Poi}(\lambda)$. Recall $\mathbb{E}[X] = \lambda$. We have

$$\begin{aligned} \mathbb{E}[X(X-1)] &= \sum_{k=2}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 \end{aligned}$$

So $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Definition 4.20. Let X and Y be random variables. Then the *covariance* of X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

We have some properties:

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
- $\text{Cov}(X, X) = \text{Var}(X)$;
- for any $c \in \mathbb{R}$, $\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$ and $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$;
- for any $c \in \mathbb{R}$, $\text{Cov}(c, X) = 0$.

Lemma 4.21

The covariance of X and Y is given by

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Proof. Expand. ■

Lemma 4.22

For random variables X and Y , we have

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

Proof. We have

$$\begin{aligned}\mathrm{Var}(X + Y) &= \mathbb{E}[(X - \mathbb{E}[X]) + (Y - \mathbb{E}[Y])]^2 \\ &= \mathrm{Var}(X) + 2 \mathrm{Cov}(X, Y) + \mathrm{Var}(Y)\end{aligned}$$

■

Lemma 4.23

For $c_1, \dots, c_n$ and $d_1, \dots, d_n \in \mathbb{R}$,

$$\mathrm{Cov}\left(\sum_{i=1}^n c_i X_i, \sum_{j=1}^n d_j Y_j\right) = \sum_{i,j=1}^n c_i d_j \mathrm{Cov}(X_i, Y_j)$$

Corollary 4.24

We have

$$\begin{aligned}\mathrm{Var}\left(\sum_{i=1}^n X_i\right) &= \mathrm{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j)\end{aligned}$$

Let us consider some results on independence.

Lemma 4.25

Suppose X_1, X_2 and X_3 are independent – then X_1 and X_2 are independent.

Proof. We have

$$\begin{aligned}\mathbb{P}(X_1 = x_1, X_2 = x_2) &= \sum_{x_3} \mathbb{P}(X_1 = x_1, X_2 = x_2, X_3 = x_3) \\ &= \sum_{x_3} \mathbb{P}(X_1 = x_1) \mathbb{P}(X_2 = x_2) \mathbb{P}(X_3 = x_3) \\ &= \mathbb{P}(X_1 = x_1) \mathbb{P}(X_2 = x_2)\end{aligned}$$

■

Remark. The result obviously generalises to any subset of a set of independent events being independent.

Lemma 4.26

Let X and Y be two independent random variables, and let $f, g : \mathbb{R} \rightarrow \mathbb{R}_+$. Then

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$$

Proof. Let the random variable $Z = (X, Y)$, so $h(Z) = f(X)g(Y)$. Then

$$\begin{aligned}\mathbb{E}[h(Z)] &= \sum_{(x,y)} h(x,y) \mathbb{P}(Z = (x,y)) \\ &= \sum_{(x,y)} f(x)g(y) \mathbb{P}(X = x, Y = y) \\ &= \sum_{(x,y)} f(x)g(y) \mathbb{P}(X = x) \mathbb{P}(Y = y) \\ &= \mathbb{E}[f(X)] \cdot \mathbb{E}[g(Y)]\end{aligned}$$

■

Proposition 4.27

If X and Y are independent, $\text{Cov}(X, Y) = 0$.

Remark. The converse is not true.

Example 4.28

Let X_1, X_2, X_3 be independent Bernoulli random variables with $p = \frac{1}{2}$. Let $Y_1 = 2X_1 - 1$, $Y_2 = 2X_2 - 1$. Also let $Z_1 = Y_1X_3$, $Z_2 = Y_2X_3$. Then $\mathbb{E}[Y_1] = \mathbb{E}[Y_2] = 0$, and $\mathbb{E}[Z_1] = \mathbb{E}[Z_2] = 0$ by independence. Then

$$\text{Cov}(Z_1, Z_2) = \mathbb{E}[Z_1 Z_2] = \mathbb{E}[Y_1 Y_2 X_3^2]$$

But we claim Z_1 and Z_2 are not independent. Observe $\mathbb{P}(Z_1 = 0, Z_2 = 0) = \mathbb{P}(X_3 = 0) = \frac{1}{2}$. But $\mathbb{P}(Z_1 = 0) = \mathbb{P}(Z_2 = 0) = \frac{1}{2}$.

§4.5 Inequalities on random variables

Proposition 4.29 (Markov's inequality)

Let X be a non-negative random variable and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof. We have

$$\begin{aligned}X &\geq a \cdot 1(X \geq a) \\ \implies E[X] &\geq a \cdot \mathbb{P}(X \geq a)\end{aligned}$$

by taking expectations. ■

Proposition 4.30 (Chebyshev's inequality)

Let X be a random variable with $\mathbb{E}[X] < \infty$. Then, for any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof. Observe

$$\begin{aligned} \mathbb{P}(|X - \mathbb{E}[X]| \geq a) &= \mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2) \\ &\leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{a^2} && \text{by Markov} \\ &= \frac{\text{Var}(X)}{a^2} \end{aligned}$$

■

Proposition 4.31 (Cauchy-Schwarz)

Let X and Y be random variables, then

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]}$$

Proof. Assume that $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$ are finite, otherwise there is nothing to prove. Then, $|XY| \leq \frac{1}{2}(X^2 + Y^2)$, so, taking expectations, $\mathbb{E}[|XY|]$ is also finite. We may also assume $\mathbb{E}[X^2] > 0$, $\mathbb{E}[Y^2] > 0$. Let us also assume for the time being that $X, Y \geq 0$.

Let $t \in \mathbb{R}$ and consider $(X - tY)^2$. We have

$$\begin{aligned} 0 &\leq \mathbb{E}[(X - tY)^2] \\ &= \mathbb{E}[X^2] - 2t\mathbb{E}[XY] + t^2\mathbb{E}[Y^2] := f(t) \end{aligned}$$

Then $f'(t) = -2\mathbb{E}[XY] + 2t\mathbb{E}[Y^2]$, so $\min f(t)$ is achieved at $t^* = \frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}$. We hence have

$$\begin{aligned} 0 &\leq \mathbb{E}[X^2] - 2\frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} + \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} \\ \implies \mathbb{E}[XY]^2 &\leq \mathbb{E}[X^2] \cdot \mathbb{E}[Y^2] \end{aligned}$$

■

Remark. We could also have just put in our value for t^* from the beginning – the minimisation approach is purely to provide rationale.

Remark. When is equality achieved? We would need $f(t^*) = 0$, that is

$$\begin{aligned} \mathbb{E}[(X - t^*Y)^2] &= 0 \\ \implies \mathbb{P}(X = t^*Y) &= \mathbb{P}(X - t^*Y = 0) = 1 \end{aligned}$$

so X is just a multiple of Y .

Definition 4.32 (Convexity). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *convex* if, for any $x, y \in \mathbb{R}$ and any $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

Proposition 4.33 (Jensen's inequality)

Let X be a random variable, and $f : \mathbb{R} \rightarrow \mathbb{R}$ a convex function. Then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

Remark. We can remember the direction of the inequality by putting $f(x) = x^2$ and using non-negativity of the variance.

Lemma 4.34

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. Then f is the supremum of all the lines lying below it. In other words, for all $m \in \mathbb{R}$, there exist $a, b \in \mathbb{R}$ such that $f(m) = am + b$ and $f(x) \geq ax + b$ for all x .

Proof of lemma. Let $m \in \mathbb{R}$, and suppose $x < m < y$. Then $m = tx + (1 - t)y$ for some $t \in [0, 1]$, or, equivalently,

$$t(m - x) = (1 - t)(y - m) \quad (*)$$

By convexity, we have

$$\begin{aligned} f(m) &= f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) \\ \implies t(f(m) - f(x)) &\leq (1 - t)(f(y) - f(m)) \\ (*) \implies \frac{f(m) - f(x)}{m - x} &\leq \frac{f(y) - f(m)}{y - m} \end{aligned}$$

Then set

$$a = \sup_{x < m} \frac{f(m) - f(x)}{m - x}$$

so that, for any $x < m < y$,

$$\frac{f(m) - f(x)}{m - x} \leq a \leq \frac{f(y) - f(m)}{y - m}$$

Hence, by rearrangement, we conclude $f(z) \geq a(z - m) + f(m)$, for any z (depending on whether z is greater than m or less than m , we use either the left or right inequality.) ■

Proof of proposition. Let $m = \mathbb{E}[X]$. By the claim, we have $a, b \in \mathbb{R}$ such that $f(m) = am + b$ and $f(x) \geq ax + b$ for all x . We have

$$\begin{aligned} f(X) &\geq aX + b \\ \implies \mathbb{E}[f(X)] &\geq a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X]) \end{aligned}$$

■

Remark. When is equality achieved?

Suppose f convex with the extra property that, for $n = \mathbb{E}[X]$, there exist $a, b \in \mathbb{R}$ with $f(m) = am + b$ but $f(x) > ax + b$, note the strict inequality, for $x \neq m$. When should X satisfy $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$?

Since $f(X) \geq aX + b$, we have $f(X) - (aX + b) \geq 0$. Taking expectations, we have

$$\mathbb{E}[f(X)] \geq a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X])$$

Since we want $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$, we must have

$$\begin{aligned} \mathbb{E}[f(X) - (aX + b)] &= 0 \\ \implies \mathbb{P}(f(X) = aX + b) &= 1 \end{aligned}$$

but then, since $f(x) > ax + b$ for $x \neq m$, we conclude $\mathbb{P}(X = m) = 1$ (if we didn't have that extra part, we'd conclude f is just a linear function).

Lecture $n + 1$

Proposition 4.35 (AM-GM inequality)

Let $x_1, \dots, x_n > 0$. Then

$$\left(\prod_{k=1}^n x_k \right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{k=1}^n x_k$$

Lemma 4.36

Let f be a convex function. Then, for any $x_1, \dots, x_n \in \mathbb{R}$ we have

$$\frac{1}{n} \sum_{k=1}^n f(x_k) \geq f\left(\frac{1}{n} \sum_{k=1}^n x_k\right)$$

Proof of lemma. Let X be a random variable with $\mathbb{P}(X = x_i) = \frac{1}{n}$, $i = 1, \dots, n$. Then

$$\frac{1}{n} \sum_{k=1}^n f(x_k) = \sum_x f(x) \mathbb{P}(X = x) = \mathbb{E}[f(X)]$$

so Jensen's gives

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) = f\left(\frac{1}{n} \sum_{k=1}^n x_k\right)$$

as required. ■

Proof of proposition. Let $f(x) = -\log x$ and suppose $x_1, \dots, x_n > 0$. Then

$$-\frac{1}{n} \sum_{k=1}^n \log x_k \geq -\log \left(\frac{1}{n} \sum_{k=1}^n x_k \right)$$

$$\implies \left(\prod_{k=1}^n x_k \right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{k=1}^n x_k$$

as required. ■

§4.6 Joint distribution and conditional expectation

Recall if X is a discrete random variable, then the distribution of X is $(\mathbb{P}(X = x))_x$.

Definition 4.37. Let $X_1, \dots, X_n$ be discrete random variables. Their *joint distribution* is defined to be the set of values of

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

for any $x_1, \dots, x_n$.

Remark. We can recover the distribution of any one of the variables from such a joint distribution with

$$\mathbb{P}(X_1 = x_1) = \sum_{x_2, \dots, x_n} \mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

In this context the distribution of the single variable is sometimes called its *marginal* distribution.

Definition 4.38. Let X and Y be discrete random variables. The *conditional distribution* of X given $Y = y$ is

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

By the law of probability we of course have

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x \mid Y = y) \mathbb{P}(Y = y)$$

Let X and Y be independent – what is the distribution of $X + Y$? We have

$$\begin{aligned} \mathbb{P}(X + Y = z) &= \sum_y \mathbb{P}(X + Y = z, Y = y) \\ &= \sum_y \mathbb{P}(X = z - y, Y = y) \\ &= \sum_y \mathbb{P}(X = z - y) \mathbb{P}(Y = y) \end{aligned}$$

where the last line is by independence. This is of course a convolution.

Example

Let $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$, with $\lambda, \mu > 0$ and $X \perp Y$. Then

$$\begin{aligned} \mathbb{P}(X + Y = n) &= \sum_{k=0}^n \mathbb{P}(X = n - k) \mathbb{P}(Y = k) \\ &= e^{-\lambda} \frac{\lambda^{n-k}}{(n-k)!} \cdot e^{-\mu} \frac{\mu^k}{k!} \\ &= \frac{e^{-(\lambda+\mu)}}{n!} \sum_{k=0}^n \binom{n}{k} \lambda^{n-k} \mu^k \\ &= e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!} \end{aligned}$$

so in particular $X + Y \sim \text{Poi}(\lambda + \mu)$.

Let us derive conditional expectation for random variables. First, take $B \in \mathcal{F}$ to be an event with $\mathbb{P}(B) > 0$. We have

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X \cdot 1(B)]}{\mathbb{P}(B)}$$

and, if $X = 1(A)$, then

$$\mathbb{E}[1(A) \mid B] = \frac{\mathbb{E}[1(A) \cdot 1(B)]}{\mathbb{P}(B)} = \frac{\mathbb{E}[1(A \cap B)]}{\mathbb{P}(B)} = \mathbb{P}(A \mid B)$$

We can now formulate the *law of total expectation*.

Proposition 4.39 (Law of total expectation)

Let X be a random variable and (Ω_n) a partition of Ω with $\mathbb{P}(\Omega_n) > 0$ for any n . Then

$$\mathbb{E}[X] = \sum_n \mathbb{E}[X \mid \Omega_n] \mathbb{P}(\Omega_n)$$

Proof. Use indicators. We have

$$\mathbb{E}[X] = \mathbb{E}[X \cdot 1(\Omega)] = \mathbb{E}\left[X \cdot \sum_n 1(\Omega_n)\right] = \sum_n \mathbb{E}[X \mid \Omega_n] \mathbb{P}(\Omega_n)$$

■

Now, we want to condition expectation not on an event but on a particular random variable. We have

$$\begin{aligned} \mathbb{E}[X \mid Y = y] &= \frac{\mathbb{E}[X \cdot 1(Y = y)]}{\mathbb{P}(Y = y)} \\ &= \sum_x x \cdot \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \\ &= \sum_x x \cdot \mathbb{P}(X = x \mid Y = y) \end{aligned}$$

Let us now define $g(y) = \mathbb{E}[X \mid Y = y]$.

Definition 4.40. The expectation of X given Y , $\mathbb{E}[X | Y]$, is a random variable which is a function of Y . In particular,

$$\mathbb{E}[X | Y] = g(Y)$$

Example

Toss a p -coin independently n times, and let $X_i = 1(i^{\text{th}} \text{ toss is H})$. Let $Y_n = X_1 + \dots + X_n$. Let us attempt to calculate $\mathbb{E}[X_1 | Y_n]$. First, let

$$\begin{aligned} g(y) = \mathbb{E}[X_1 | Y_n = y] &= \frac{\mathbb{E}[X_1 \cdot 1(Y_n = y)]}{\mathbb{P}(Y_n = y)} \\ &= \frac{\mathbb{P}(X_1 = 1, Y_n = y)}{\mathbb{P}(Y_n = y)} \\ &= \frac{p \binom{n-1}{y-1} p^{y-1} (1-p)^{n-y}}{\binom{n}{y} p^y (1-p)^{n-y}} \\ &= \frac{y}{n} \end{aligned}$$

Hence, $\mathbb{E}[X_1 | Y_n] = \frac{Y_n}{n}$.

We have some simple and uninspiring properties:

- $\mathbb{E}[cX | Y] = c\mathbb{E}[X | Y]$;
- $\mathbb{E}[c + X | Y] = c + \mathbb{E}[X | Y]$;
- $\mathbb{E}[c | Y] = c$;
- If $X_1, \dots, X_n$ are random variables,

$$\mathbb{E} \left[\sum_{i=1}^n X_i | Y \right] = \sum_{i=1}^n \mathbb{E}[X_i | Y]$$

and some more interesting properties. We have

$$\begin{aligned} \mathbb{E}[X | Y] &= \sum_y 1(Y = y) \mathbb{E}[X | Y = y] \\ \implies \mathbb{E}[\mathbb{E}[X | Y]] &= \sum_y \mathbb{E}[X | Y = y] \mathbb{P}(Y = y) = \mathbb{E}[X] \end{aligned}$$

Suppose now X and Y are independent random variables. Then

$$\begin{aligned} \mathbb{E}[X | Y] &= \sum_y 1(Y = y) \mathbb{E}[X | Y = y] \\ &= \sum_y 1(Y = y) \sum_x x \cdot \mathbb{P}(X = x | Y = y) \\ &= \sum_y 1(Y = y) \sum_x x \cdot \mathbb{P}(X = x) \\ &= \sum_y 1(Y = y) \mathbb{E}[X] = \mathbb{E}[X] \end{aligned}$$

Proposition 4.41

Let Y and Z be independent. Then

$$\mathbb{E}[\mathbb{E}[X | Y] | Z] = \mathbb{E}[X]$$

Proof. Let $g(Y) = \mathbb{E}[X | Y]$, $g(y) = \mathbb{E}[X | Y = y]$. We claim $g(Y)$ is independent of Z . Observe

$$\begin{aligned} \mathbb{P}(g(Y) = w, Z = z) &= \sum_{y \in g^{-1}(\{w\})} \mathbb{P}(Y = y, Z = z) \\ &= \sum_{y \in g^{-1}(\{w\})} \mathbb{P}(Y = y) \mathbb{P}(Z = z) \\ &= \mathbb{P}(g(Y) = w) \mathbb{P}(Z = z) \end{aligned}$$

Hence, previous results give $\mathbb{E}[g(Y) | Z] = \mathbb{E}[g(Y)]$, and this is $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$. ■

Proposition 4.42

Let X, Y be two random variables, and $h : \mathbb{R} \rightarrow \mathbb{R}$. Then

$$\mathbb{E}[h(Y) \cdot X | Y] = h(Y) \cdot \mathbb{E}[X | Y]$$

Proof. We have

$$\begin{aligned} \mathbb{E}[h(Y) \cdot X | Y] &= \sum_y 1(Y = y) \mathbb{E}[h(Y) \cdot X | Y = y] \\ &= \sum_y 1(Y = y) h(y) \mathbb{E}[X | Y = y] \\ &= h(Y) \sum_y 1(Y = y) \mathbb{E}[X | Y = y] \\ &= h(Y) \mathbb{E}[X | Y] \end{aligned}$$

■

Corollary 4.43

$$\mathbb{E}[\mathbb{E}[X | Y] | Y] = \mathbb{E}[X | Y]$$

Example

Toss a p -coin independently n times, and let $X_i = 1$ (i^{th} toss is H). Let $Y_n = X_1 + \cdots + X_n$. What is $\mathbb{E}[X_1 | Y_n]$?

By symmetry, $\mathbb{E}[X_1 | Y_n] = \mathbb{E}[X_i | Y_n]$ for any i , $1 \leq i \leq n$. Then

$$Y_n = \mathbb{E}[Y_n | Y_n] = \mathbb{E}[X_1 + \cdots + X_n | Y_n] = n \mathbb{E}[X_1 | Y_n]$$

which was what we wanted.

§5 Random walks

Definition 5.1 (Random process). A *random process* or *stochastic process* is a sequence of random variables (X_n) .

Definition 5.2 (Random walks). A *random walk* is a random process that can be expressed as

$$X_n = x + Y_1 + \cdots + Y_n$$

where x is a number and (Y_i) are independent identically distributed (i.i.d.) random variables. We call the Y_i the *steps* of the random variable.

A *simple random walk* has

$$Y_i = \begin{cases} 1 & \text{with probability } p \\ -1 & \text{with probability } q = 1 - p \end{cases}$$

We will write

- $\mathbb{P}_x(A) = \mathbb{P}(A \mid X_0 = x)$;
- $h(x) = \mathbb{P}_x((X_n) \text{ reaches } a \text{ before } 0)$, where a is some arbitrary constant at which the game ends.

so $h(0) = 0$, $h(a) = 1$. Then

$$\begin{aligned} h(x) &= \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0) \\ &= \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0 \mid Y_1 = 1) \cdot p + \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0 \mid Y_1 = -1) \cdot q \\ &= p \cdot h(x+1) + q \cdot h(x-1) \end{aligned}$$

Lecture 12 – 18/02/26

- Case $p = q = \frac{1}{2}$, then X_n is a *simple symmetric random walk*. We have

$$\begin{aligned} h(x) &= \frac{1}{2}h(x+1) + \frac{1}{2}h(x-1) \\ \implies h(x+1) - h(x) &= h(x) - h(x-1) \\ \implies h(x) &= cx \end{aligned}$$

for some constant c . Since $h(a) = 1$, $c = \frac{1}{a}$, and hence $h(x) = \frac{x}{a}$.

- Case $p \neq q$. Let us try a solution of the form λ^x for some λ – we have

$$\begin{aligned} \lambda^x &= p\lambda^{x+1} + q\lambda^{x-1} \\ \implies \lambda &= 1, \lambda = \frac{q}{p} \end{aligned}$$

so the general solution has the form

$$h(x) = A + B \left(\frac{q}{p} \right)^x$$

which, applying boundary conditions, gives

$$h(x) = \frac{\left(\frac{q}{p}\right)^x - 1}{\left(\frac{q}{p}\right)^a - 1}$$

These results are called the *gambler's ruin estimates*.

Now let us consider a new quantity – let $T = \min\{n \geq 0 : X_n \in \{0, a\}\}$ – the first n such that X_n is either 0 or a , i.e. the duration of the game. We want to find $\mathbb{E}[T \mid X_0 = x] = \mathbb{E}_x[T] = k(x)$. We have from the law of total expectation

$$\begin{aligned} k(x) &= \mathbb{E}_x[T \mid Y_1 = 1] \cdot \mathbb{P}(Y_1 = 1) + \mathbb{E}_x[T \mid Y_1 = -1] \cdot \mathbb{P}(Y_1 = -1) \\ &= p(1 + k(x+1)) + q(1 + k(x-1)) \\ &= 1 + pk(x+1) + qk(x-1) \end{aligned}$$

with $k(0) = k(a) = 0$.

- Case $p = q = \frac{1}{2}$. Try a solution of the form Ax^2 , to get

$$\begin{aligned} Ax^2 &= 1 + pA(x+1)^2 + qA(x-1)^2 \\ \implies A &= -1 \end{aligned}$$

giving a general solution of the form $k(x) = -x^2 + Bx + C$. Applying boundary conditions, we obtain

$$k(x) = x(a-x)$$

- Case $p \neq q$. Try $k(x) = Cx$ to get

$$\begin{aligned} Cx &= 1 + pC(x+1) + qC(x-1) \\ \implies c &= \frac{1}{q-p} \end{aligned}$$

Adding this to the solution of the homogeneous equation, we obtain

$$k(x) = A + \frac{1}{q-p}x + B\left(\frac{q}{p}\right)^x$$

which, after boundary conditions, gives

$$k(x) = \frac{1}{q-p}x - \frac{a}{q-p} \cdot \frac{\left(\frac{q}{p}\right)^x - 1}{\left(\frac{q}{p}\right)^a - 1}$$

§6 Probability generating functions

Let X be a random variable with values in $\mathbb{N}$ – recall $p_r = \mathbb{P}(X = r)$ for $r \in \mathbb{N}$ constitutes the probability mass function.

Definition 6.1 (Probability generating function). The *probability generating function (PGF)* for a random variable X with values in $\mathbb{N}$ and probability masses (p_r) is

$$p(z) = \mathbb{E}[z^X] = \sum_{r=0}^{\infty} p_r z^r$$

with $|z| \leq 1$.

Remark. This is well-defined because

$$|p(z)| = \left| \sum_{r=0}^{\infty} p_r z^r \right| \leq \sum_{r=0}^{\infty} p_r |z|^r \leq \sum_{r=0}^{\infty} p_r = 1$$

so p converges absolutely for $|z| \leq 1$.

Theorem 6.2

The pgf uniquely determines the distribution of X .

Proof. Let (p_r) and (q_r) be two probability distributions with the same pgf's – for all $|z| \leq 1$, $\sum p_r z^r = \sum q_r z^r$.

Plugging in $z = 0$ gives $p_0 = q_0$. We now proceed by strong induction. Suppose $p_r = q_r$ for all $r \leq n$, so

$$\sum_{r=n+1}^{\infty} p_r z^r = \sum_{r=n+1}^{\infty} q_r z^r$$

Clearly dividing by z^{n+1} and setting $z = 0$ gives $p_{n+1} = q_{n+1}$ as required. ■

Proposition 6.3

Suppose $X_1, \dots, X_n$ are independent random variables with pgf's $q_i(z)$, and let $X = X_1 + \dots + X_n$ have pgf p . Then

$$p(z) = q_1(z) \dots q_n(z)$$

Proof. Clearly

$$p(z) = \mathbb{E}[z^{X_1 + \dots + X_n}] = \mathbb{E}[X_1] \dots \mathbb{E}[X_n] = q_1(z) \dots q_n(z)$$

by independence. ■

Example

Let us calculate some pgf's for important distributions.

- ① *Binomial*. Suppose $X \sim \text{Bin}(n, p)$. Then

$$\begin{aligned} p(z) &= \sum_{k=0}^n z^k \binom{n}{k} p^k (1-p)^{n-k} \\ &= (pz + 1 - p)^n \end{aligned}$$

- ② *Geometric*. Suppose $X \sim \text{Geo}(p)$. Then

$$\begin{aligned} \mathbb{E}[z^X] &= \sum_{r=0}^{\infty} z^r (1-p)^r p \\ &= \frac{p}{1 - z(1-p)} \end{aligned}$$

- ③ *Poisson*. Suppose $X \sim \text{Poi}(\lambda)$. Then

$$\begin{aligned} \mathbb{E}[z^X] &= \sum_{r=0}^{\infty} z^r e^{-\lambda} \frac{\lambda^r}{r!} \\ &= e^{\lambda(z-1)} \end{aligned}$$

Example

This fact can make calculating the pmf of sums of random variables easier. For example, suppose $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ with $X \perp Y$. Then

$$\begin{aligned} \mathbb{E}[z^{X+Y}] &= \mathbb{E}[z^X] \mathbb{E}[z^Y] = (pz + 1 - p)^n \cdot (pz + 1 - p)^m \\ &= (pz + 1 - p)^{n+m} \end{aligned}$$

from which it follows that $X + Y \sim \text{Bin}(n + m, p)$.

What about Poisson? Suppose $X \sim \text{Poi}(\lambda)$, $Y \sim \text{Poi}(\mu)$, and $X \perp Y$. Then

$$\mathbb{E}[z^{X+Y}] = e^{\lambda(z-1)} e^{\mu(z-1)} = e^{(\lambda+\mu)(z-1)}$$

from which it follows that $X + Y \sim \text{Poi}(\lambda + \mu)$.

For the following results, it will be helpful to introduce the notation

$$p(1^-) = \lim_{z \rightarrow 1^-} p(z)$$

Proposition 6.4

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{E}[X] = p'(1^-)$$

Proof. • Assume $\mathbb{E}[X] < \infty$, and take $0 < z < 1$. We have

$$p'(z) = \sum_{r=1}^{\infty} r p_r z^{r-1} \leq \sum_{r=1}^{\infty} r p_r = \mathbb{E}[X]$$

Hence p' is upper bounded by $\mathbb{E}[X]$, and it is clearly increasing, so the limit exists and is $\leq \mathbb{E}[X]$.

Let $\varepsilon > 0$. There exists N such that

$$\sum_{r=1}^N r p_r \geq \mathbb{E}[X] - \varepsilon$$

since this is what it means for the infinite sum to be equal to $\mathbb{E}[X]$. Furthermore,

$$\lim_{z \rightarrow 1^-} p'(z) \geq \lim_{z \rightarrow 1^-} \sum_{r=1}^N r p_r z^{r-1} = \sum_{r=1}^N r p_r \geq \mathbb{E}[X] - \varepsilon$$

• If $\mathbb{E}[X] = \infty$, then, for any $M \geq 0$, we have N such that

$$\sum_{r=1}^N r p_r \geq M$$

Then, as above,

$$\lim_{z \rightarrow 1^-} p'(z) \geq \lim_{z \rightarrow 1^-} \sum_{r=1}^N r p_r z^{r-1} = \sum_{r=1}^N r p_r \geq M$$

so the limit is infinite. ■

Proposition 6.5

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{E}[X(X-1)\dots(X-k+1)] = p^{(k)}(1^-)$$

Corollary 6.6

Let X be a discrete random variable with pgf $p(z)$, then

$$\text{Var}(X) = p''(1^-) + p'(1^-) - (p'(1^-))^2$$

Proposition 6.7

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{P}(X = n) = \frac{1}{n!} \left. \frac{d^n p}{dz^n} \right|_{z=0}$$

Remark. This is another way to see [theorem 6.2](#).

§6.1 Sum of a random number of random variables

Let $X_1, X_2, \dots$ be i.i.d. random variables and let N be an independent random variable with values in $\mathbb{N}$. Set

$$S_n = X_1 + \dots + X_n$$

that, is

$$S_n(\omega) = \sum_{i=1}^n X_i(\omega)$$

Passing to the random variable N , we obtain

$$S_N(\omega) = S_{N(\omega)}(\omega) = \sum_{i=1}^{N(\omega)} X_i(\omega)$$

Lemma 6.8

Let $q(z) = \mathbb{E}[z^N]$ be the pgf of N , and $p(z) = \mathbb{E}[z^{X_i}]$ be the pgf of X_i . Then

$$r(z) = \mathbb{E}[z^{S_N}] = q(p(z))$$

Proof 1. We have

$$\begin{aligned} r(z) &= \mathbb{E}[z^{S_N}] = \sum_{n=0}^{\infty} \mathbb{E}[z^{S_N} \cdot 1(N = n)] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[z^{S_n} \cdot 1(N = n)] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[z^{X_1 + \dots + X_n} \cdot 1(N = n)] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[z^{X_1 + \dots + X_n}] \mathbb{P}(N = n) \\ &= \sum_{n=0}^{\infty} (p(z))^n \mathbb{P}(N = n) \\ &= q(p(z)) \end{aligned}$$

■

Proof 2. We have

$$r(z) = \mathbb{E}[z^{S_N}] = \mathbb{E}[\mathbb{E}[z^{S_N} \mid N]] = g(N)$$

where

$$g(n) = \mathbb{E}[z^{S_N} \mid N = n]$$

$$\begin{aligned} &= \mathbb{E}[z^{S_n}] \\ &= p(z)^n \end{aligned}$$

so that

$$\begin{aligned} r(z) &= \mathbb{E}[p(z)^n] \\ &= q(p(z)) \end{aligned}$$

■

Corollary 6.9

We have

$$\begin{aligned} \mathbb{E}[S_n] &= \mathbb{E}[N] \cdot \mathbb{E}[X_1] \\ \text{Var}(S_n) &= \mathbb{E}[N] \cdot \text{Var}(X_1) + \text{Var}(N) \cdot \mathbb{E}[X_1]^2 \end{aligned}$$

Proof. The first result follows from

$$\begin{aligned} \mathbb{E}[S_n] &= r'(1-) \\ &= \lim_{z \rightarrow 1-} r'(z) \\ &= \lim_{z \rightarrow 1-} q'(p(z)) \lim_{z \rightarrow 1-} p'(z) \end{aligned}$$

Clearly $p(z)$ approaches 1 from below as z approaches 1 from below, so this is just

$$q'(1-)p'(1-) = \mathbb{E}[N] \cdot \mathbb{E}[X_1]$$

as required.

The second result is left as an exercise.

■

§7 Branching processes

Think of a branching process as a series of generations of offspring. We have $X_0 = 1$, there being one person in the first generation; then, in each generation, each individual produces k offspring independently with a fixed distribution (in particular, the distribution of X_1)

Let $Y_{k,n}$ with $k \geq 1$, $n \geq 0$ be i.i.d. random variables with the distribution of X_1 , interpreted as the number of offspring produced by the k^{th} individual in the n^{th} generation. Hence

$$X_{n+1} = \begin{cases} Y_{1,n} + \cdots + Y_{X_n,n} & X_n \geq 1 \\ 0 & X_n = 0 \end{cases}$$

Proposition 7.1

We have

$$\mathbb{E}[X_n] = \mathbb{E}[X_1]^n$$

Proof. We will proceed by induction. The base case is clear. Let us consider the conditional expectation $\mathbb{E}[X_{n+1} \mid X_n = m]$ – we have

$$\begin{aligned}\mathbb{E}[X_{n+1} \mid X_n = m] &= \mathbb{E}[Y_{1,n} + \cdots + Y_{m,n} \mid X_n = m] \\ &= \mathbb{E}[Y_{1,n} + \cdots + Y_{m,n}] && \text{(independence)} \\ &= m\mathbb{E}[X_1]\end{aligned}$$

so that $\mathbb{E}[X_{n+1} \mid X_n] = X_n\mathbb{E}[X_1]$. Hence $\mathbb{E}[X_{n+1}] = \mathbb{E}[\mathbb{E}[X_{n+1} \mid X_n]] = \mathbb{E}[X_n\mathbb{E}[X_1]] = \mathbb{E}[X_n]\mathbb{E}[X_1]$, which completes the result by induction. ■

Proposition 7.2

Let $G(z) = \mathbb{E}[z^{X_1}]$, $G_n(z) = \mathbb{E}[z^{X_n}]$, then

$$G_{n+1}(z) = G(G_n(z))$$

Proof. Again, let us consider $\mathbb{E}[z^{X_{n+1}} \mid X_n = m]$. We have

$$\begin{aligned}\mathbb{E}[z^{X_{n+1}} \mid X_n = m] &= \mathbb{E}[z^{Y_{1,n} + \cdots + Y_{m,n}} \mid X_n = m] \\ &= \mathbb{E}[z^{Y_{1,n} + \cdots + Y_{m,n}}] \\ &= (G(z))^m\end{aligned}$$

Hence $G_{n+1}(z) = \mathbb{E}[\mathbb{E}[z^{X_{n+1}} \mid X_n]] = \mathbb{E}[G(z)^{X_n}] = G_n(G(z))$ ■

§7.1 Extinction probability

Let $q_n = \mathbb{P}(X_n = 0)$, and $q = \mathbb{P}(X_n = 0 \text{ for some } n \geq 0)$. Letting A_n be the event that $X_n = 0$ for some n , we can see $q = \mathbb{P}(\cup A_n)$, and $A_n \subseteq A_{n+1}$, so the A_n form an increasing sequence and we get

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcup A_n\right)$$

by [proposition 1.8](#) (continuity of probability measures).

Lemma 7.3

Let $G(z) = \mathbb{E}[z^{X_1}]$, then $q_{n+1} = G(q_n)$ and $q = G(q)$.

Proof 1. We have

$$\begin{aligned}q_{n+1} &= \mathbb{P}(X_{n+1} = 0) \\ &= G_{n+1}(0) \\ &= G(G_n(0)) \\ &= G(q_n)\end{aligned}$$

Then, since G is continuous and $q_n \rightarrow q$, we can simply take the limit on both sides to get $q = G(q)$ as required. ■

Proof 2. We will condition q_{n+1} on the number of individuals in the first generation, i.e. X_1 . Supposing $X_1 = m$, let $X_n^{(1)}, \dots, X_n^{(m)}$ be i.i.d. branching processes with the same distribution as the original one. Hence

$$X_{n+1} = X_n^{(1)} + \dots + X_n^{(m)}$$

So

$$\begin{aligned} q_{n+1} &= \mathbb{P}(X_{n+1} = 0) \\ &= \sum_m \mathbb{P}(X_{n+1} = 0 \mid X_1 = m) \mathbb{P}(X_1 = m) \\ &= \sum_m \mathbb{P}(X_n^{(1)} + \dots + X_n^{(m)} = 0 \mid X_1 = m) \mathbb{P}(X_1 = m) \\ &= \sum_m \left(\mathbb{P}(X_n^{(1)} = 0) \right)^m \mathbb{P}(X_1 = m) \\ &= \sum_m q_n^m \mathbb{P}(X_1 = m) \\ &= G(q_n) \end{aligned}$$

Once again, we take the limit on both sides. ■

Lecture 14 – 23/02/26

Recalling [proposition 7.1](#), we have that

- if $\mathbb{E}[X_1] < 1$, $\mathbb{E}[X_n] \rightarrow 0$ as $n \rightarrow \infty$, so we expect $q = 1$;
- if $\mathbb{E}[X_1] > 1$, $\mathbb{E}[X_n] \rightarrow \infty$ as $n \rightarrow \infty$, so we expect $q < 1$;
- if $\mathbb{E}[X_1] = 1$, then $\mathbb{E}[X_n] = 1$ for all n . But it is not clear what q will be. Let us focus on this last case.

We can also interpret these results in the context of a graph of G . We know that the graph of G will intersect x at $x = 1$, and we could either have another intersection in the range $0 \leq x < 1$, or no intersections.

- *There is another intersection.* Then iterating G starting from $G(0)$ will lead us to the other intersection. Furthermore, the gradient of G at 1, $G(1-) = \mathbb{E}[X_1]$ must be greater than 1 for there to be such an intersection, as we expect;
- *There is not another intersection.* Then iterating G will take us all the way to 1, and the gradient of G at 1, $G(1-) = \mathbb{E}[X_1]$ must be less than or equal to 1 for there not to be any further intersections. We conclude that $q = 1$ in the case $\mathbb{E}[X_1] = 1$.

Proposition 7.4

Assume $P(X_1 = 1) < 1$. The extinction probability q is the minimal non-negative solution to $G(x) = x$. Furthermore, $q < 1$ if and only if $\mathbb{E}[X_1] > 1$.

Proof. We know $q = G(q)$. Let t be the smallest minimal solution of $x = G(x)$. We will show by induction that, for any n , $q_n \leq t$, and then take limits to establish $q \leq t \implies q = t$.

For the base case, $0 = q_0 \leq t$. Suppose $q_n \leq t$, then

$$q_{n+1} = G(q_n) \leq G(t) = t$$

since G is increasing.

Now, we show $q < 1$ if and only if $\mathbb{E}[X_1] > 1$. Suppose for now $g_0 + g_1 = \mathbb{P}(X_1) \leq 1 < 1$.

Let $H(z) = G(z) - z = \sum_{r=0}^{\infty} g_r z^r - z$, so $H(1) = 0$. First, we show that H can have at most one root in $[0, 1)$. Observe

$$H''(z) = \sum_{r=0}^{\infty} r(r-1)g_r z^{r-1} > 0$$

in $(0, 1)$, because $g_0 + g_1 < 1$, so there must be some later r with a positive value of g_r , making the sum strictly larger than zero. Hence $H'(z)$ is strictly increasing, and hence can have at most one more root other than 1 – otherwise, we would have a point c with $H'(c) = 0$ by Rolle's theorem.

We now split into two cases.

- H has no root other than 1, so $q = 1$. Then $H'(1-) = G'(1-) - 1 = \mathbb{E}[X_1] - 1$. Furthermore, $H(0) = G(0) = g_0 \geq 0$ while $H(1) = 0$, so $H(z) \geq 0$ for all $z \in [0, 1]$. Now

$$H'(1-) = \lim_{z \rightarrow 1-} \frac{H(z) - H(0)}{z - 1} \leq 0$$

since the numerator is $H(z) - 0 \geq 0$ while the denominator is $z - 1 \leq 0$. Hence $H'(1-) \leq 0 \implies \mathbb{E}[X_1] \leq 1$, as required.

- H has one more root r , so r is the extinction probability. We hence have $H(r) = 0$ and $H(1) = 0$ with $r < 1$, so we have $z \in (r, 1)$ with $H'(z) = 0$ by Rolle. Hence $\mathbb{E}[X_1] - 1 = H'(-1) > H'(z) = 0$, so $\mathbb{E}[X_1] > 1$ as required.

On the other hand, if $\mathbb{P}(X_1 \leq 1) = 1$, then $\mathbb{E}[X_1] = g_1$, so

$$\begin{aligned} G(z) &= g_0 + g_1 z \\ &= 1 - \mathbb{E}[X_1] + \mathbb{E}[X_1]z \end{aligned}$$

Then, if $G(z) = z$, we have $(1 - \mathbb{E}[X_1])z = 1 - \mathbb{E}[X_1]$ and, since $\mathbb{E}[X_1] = g_1 < 1$, $1 - \mathbb{E}[X_1] \neq 0$ so $z = 1$. ■

§8 Continuous random variables

Recall [definition 4.2](#), that

Definition. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \rightarrow \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for any x .

and [definition 4.4](#), that

Definition. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned} F_X(x) : \mathbb{R} &\rightarrow [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(X \leq x) \end{aligned}$$

When dealing with discrete random variables, it was possible to only consider $\mathbb{P}(X = x)$ for some x . But now, we will have to work in more generality.

Let us observe some important properties.

- ① if $x \leq y$, then $F(x) \leq F(y)$;
- ② for any $a < b$,

$$\mathbb{P}(a < x \leq b) = F(b) - F(a)$$